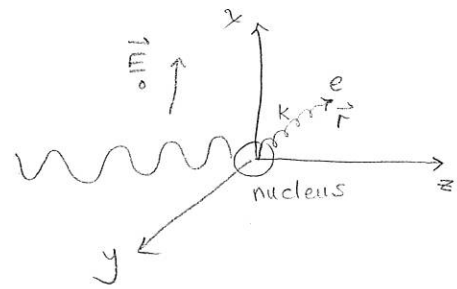
Feb 6, 201

Classical Model of atom:

Electron in an atom, is bounded to nucleus

This is Problem of harmonic oscillator in an external Field

$\vec{E} = \vec{E}_0 \cos \omega t$



Q: How much radiation this charge is being scattered in solid angle?

Consider: (Size of atom) $a \ll \lambda$

Set up Newton's Law: $\vec{F} = m\ddot{\vec{r}} = -k\vec{r} + eE_0 \cos(\omega t) + \frac{\mu_0 e^2}{6\pi c}\dddot{\vec{r}}$ $\xrightarrow{k = m\omega_0^2}$

$m\ddot{\vec{r}} = -m\omega_0^2 \vec{r} + e\,\mathrm{Re}(E_0 e^{-i\omega t}) + \frac{\mu_0 e^2}{6\pi c}\dddot{\vec{r}}$: just think about inhomogeneous solution:

try $\vec{r} = \vec{r}_0 e^{-i\omega t}$ (Since, in long time, each driven oscillator, is derived, just by external freq

Substitute in above equation:

$$-m\omega^2 \vec{r}_0 \cancel{e^{-i\omega t}} = -m\omega_0^2 \vec{r}_0 \cancel{e^{-i\omega t}} + e\,(E_0 e^{-i\omega t}) + \frac{i\mu_0 e^2}{6\pi c}\omega^3 \vec{r}_0 \cancel{e^{-i\omega t}} \Rightarrow$$

$$\vec{r}_0 = \frac{e\vec{E}_0}{m(\omega_0^2 - \omega^2 - i\omega\gamma(\omega))}, \quad \gamma(\omega) = \frac{\mu_0 e^2 \omega^2}{6\pi m c}$$

$\vec{r} = \vec{r}_0 e^{-i\omega t}$: $\vec{r} = \dfrac{e\vec{E}_0 e^{-i\omega t}}{m(\omega_0^2 - \omega^2 - i\omega\gamma\omega)}$

$\vec{d} = e\vec{r}$, dipole moment is given by real part : $\begin{cases} ① \text{ multiply by complex conjugate of denominator} \\ ② \; e^{-i\omega t} = \cos\omega t - i\sin\omega t \end{cases}$

$$\vec{d} = \frac{e^2\left((\omega_0^2-\omega^2)\cos\omega t + \gamma\omega\sin\omega t\right)\vec{E}_0}{m\left((\omega_0^2-\omega^2)^2 + \gamma^2\omega^2\right)}$$

wave vector is along $\hat{z}$ and E is along $\hat{x}$, Since $\vec{d}$ is harmonic function,

we can get readily $\dot{\vec{d}}$, $\ddot{\vec{d}}$:

$$\dot{\vec{d}} = \frac{e^2\vec{E}_0\left(-\omega(\omega_0^2-\omega^2)\sin\omega t + \gamma\omega^2\cos\omega t\right)}{m\left((\omega_0^2-\omega^2)^2 + \gamma^2\omega^2\right)}$$

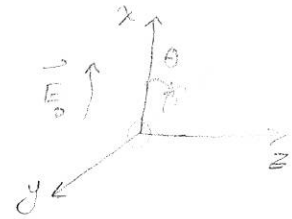
$$\ddot{\vec{d}} = \frac{e^2\vec{E}_0\left(-\omega^2(\omega_0^2-\omega^2)\cos\omega t - \gamma\omega^3\sin\omega t\right)}{m\left((\omega_0^2-\omega^2)^2 + \gamma^2\omega^2\right)}$$

$$|\ddot{\vec{d}}|^2 = \left(\frac{e^2E_0}{m((\omega_0^2-\omega^2)^2+\gamma^2\omega^2)}\right)^2\left(\omega^4(\omega_0^2-\omega^2)^2\cos^2\omega t + \gamma^2\omega^6\sin^2\omega t + 2\gamma\omega^3(\omega_0^2-\omega^2)\cos\omega t\sin\omega t\right)$$

take time avg : $\langle\cos^2\omega t\rangle_t = \langle\sin^2\omega t\rangle_t = \frac{1}{2}$, $\langle\sin\omega t\cos\omega t\rangle_t = 0$

$$\langle\ddot{\vec{d}}^2\rangle = \left(\frac{e^2E_0}{m}\right)^2\left(\frac{\omega^4}{2}\right)\left(\frac{1}{(\omega_0^2-\omega^2)^2+\gamma^2\omega^2}\right)$$

dipole Intensity: $\frac{dI}{d\Omega} = \frac{\mu_0}{16\pi^2 c}(\ddot{\vec{d}} \times \hat{n})^2 = \frac{\mu_0}{16\pi^2 c}\ddot{d}^2 \sin^2\theta$



direction of polarization and scattering, only matters:

Intensity: $I = \int \frac{\mu_0}{16\pi^2 c}\ddot{d}^2 \sin^2\theta \, d\Omega$, where $\int \sin^2\theta \, d\Omega = \frac{2}{3}\cdot 4\pi = \frac{8\pi}{3}$

$$P(=I) = \frac{\mu_0}{12\pi c}\left(\frac{e^2 E_0 \omega^2}{m}\right)^2 \left(\frac{1}{(\omega_0^2-\omega^2)^2+\gamma^2\omega^2}\right)$$

Radiated Power is proportional to incident Electric Field, but we want to define some quantity, to doesn't depend to E_0:

define, scattering cross section of area: $\sigma = \frac{P}{\langle|\vec{S}|\rangle}$

energy flux: $|\vec{S}| = \frac{1}{\mu_0}|\vec{E}\times\vec{B}| \xrightarrow[c^2 = \frac{1}{\varepsilon_0\mu_0}]{B = E/c} |\vec{S}| = \varepsilon_0 c E^2$

energy density: $W = \varepsilon_0 E^2$: $S = W \cdot c = \varepsilon_0 c E_0^2 \cos^2(\omega t)$

$\langle|\vec{S}|\rangle = \frac{1}{2} c \varepsilon_0 E_0^2$

$$\sigma = \frac{P}{\langle|\vec{S}|\rangle} = \frac{e^4\omega^4}{6\pi c^4 m^2 \varepsilon_0^2 ((\omega_0^2-\omega^2)^2+\gamma^2\omega^2)}$$

Potential energy : $\frac{e^2}{4\pi\varepsilon_0 R_0} = mc^2$: energy

Classical electron radius : $R_0 = \frac{e^2}{4\pi m\varepsilon_0 c^2}$, Simplify σ in terms of R_0 :

$$\sigma = \frac{8\pi}{3} R_0^2 \frac{\omega^4}{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2}$$

Now, analyze in different limits (for all frequencies γ is much much smaller than ω) :

① $\omega \ll \omega_0$ → $\gamma\omega$ is very small → neglect $\gamma\omega$

$$\sigma = \frac{8\pi}{3} R_0^2 \omega^4 \frac{1}{\omega_0^4 (1 - (\frac{\omega}{\omega_0})^2)^2 + \cancel{\gamma^2\omega^2}} \Rightarrow \sigma = \frac{8\pi}{3} R_0^2 \left(\frac{\omega}{\omega_0}\right)^4$$ → "the blue sky" Formula

② $\omega \approx \omega_0$:

$$\sigma = \frac{8\pi}{3\gamma^2} R_0^2 \omega_0^2$$

* gamma (γ) determines both width and height of peak . Smaller γ → higher peak

(3) $\omega \gg \omega_0$, γ is Small, even we are close to the resonance $\rightarrow$ neglect γ

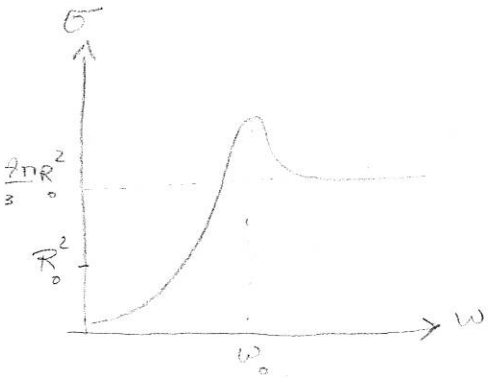
$$\sigma = \frac{8\pi}{3} R_0^2 \rightarrow \text{Constant}$$, doesn't depend on ω, ω_0

Physical meaning

$\downarrow$

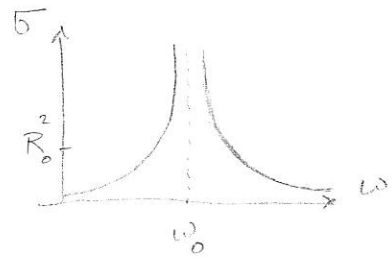
It's vibrating so fast (X-ray Scattering).

* all of this, only works, if friction radiation is Small.

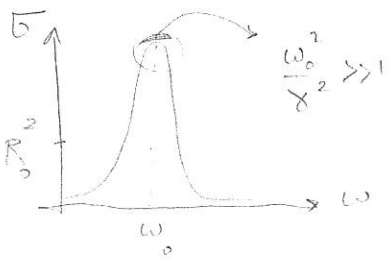


* Since γ is so Small, Max value happens for length Scale, greater than R_0^2

* If have no γ, we have Singularity in denominator : $\sigma = \frac{8\pi}{3} R_0^2 \frac{\omega^4}{(\omega_0^2 - \omega^2)^2}$



* Think of γ, just alone : $\sigma = \frac{8\pi}{3} R_0^2 \left(\frac{\omega^2}{\gamma^2}\right)$



* Since γ is very Small, Causes the Curve not to go to infinity.

* Look at γ term :

If γ exceeds ω : $\gamma_{(\omega)} = \frac{\mu_0 e^2 \omega^2}{6\pi m c} \gg \omega$, $\mu_0 = \frac{1}{\varepsilon_0 c^2}$, $R_0 = \frac{e^2}{4\pi m \varepsilon_0 c^2}$

$$\left(\frac{e^2}{6\pi m \varepsilon_0 c^2}\right)\left(\frac{\omega}{c}\right) \gg 1 \xrightarrow{\frac{\omega}{c} \propto \frac{1}{\lambda}} \simeq \frac{R_0}{\lambda} \gg 1 \; : \; R_0 \gg \lambda$$

which doesn't work and violates Quantum mechanics